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2026-09-21

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Honor Pledge: I pledge that the only resources I used to complete this homework were the course textbook, the course slides, or my personal course notes. I did not confer with any other student, consult any solutions manuals, printed or on-line, nor did I do any Internet searches to solve any problems for this homework assignment. The solutions below are my own work.

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Cyber Security Fundamentals - CHAPTER 6 HOMEWORK

70 points total

Problem 6.1 [15] There are many weaknesses in the virus program below (i.e., trivial to detect via signature, increases size of executables, etc.) The logic of the program is fine too. However, there is a significant operational flaw. What is that flaw?

Hint: Consider what happens towards the end of the infection cycle.

```
program V
1234567;

procedure attach-to-program;
begin
  repeat
    file := get-random-program;
    until first-program-line ≠ 1234567;
    prepend V to file;
  end;

procedure execute-payload;
begin
  (* perform payload actions *)
end;

procedure trigger-condition;
begin
  (* return true if trigger condition is true *)
end;

begin (* main action block *)
  attach-to-program;
  if trigger-condition then execute-payload;
  goto main;
end;
```

Solution:

The major flaw in the program above is that it has an infinite loop. In the procedure attach-to-program, it will repeatedly cycle through random files until it can infect all of them with 1234567. Then the statement: “until first-program-line ≠ 1234567” will never equal true, causing an infinite loop that will max out the CPU and crash or freeze the process.

- 6.3.** [20] The following code fragments show a sequence of virus instructions and a metamorphic version of the virus. Describe the effect produced by the metamorphic code.

Original Code	Metamorphic Code
move eax, 5 add eax, ebx call [eax]	mov eax, 5 push ecx pop ecx add eax, ebx swap eax, ebx swap ebx, eax call [eax] nop

Brief explanation of above assembly language code:

eax, ebx, ecx, are the names of *registers* - essentially single word memory storage locations. Numeric values are stored in registers.

move eax, 5 ; moves the value '5' into the eax register

add eax, ebx; adds the values in eax and ebx and stores the result in eax.

 ; if eax is storing the value of 5 and ebx is storing 3, after the instruction

 ; executes, eax will have 8 (the result of the addition) and ebx will still contain

 ; the value 3

call [eax] ; starts executing code at memory location specified in eax, in this case 8

push ecx ; writes the value in ecx to the 'stack' (don't worry about what the stack is, its

 ; just another area of memory)

pop ecx ; takes the last value written to the stack (i.e., push ecx) and writes the value

 ; into ecx

swap eax, ebx ; the value that was in eax is written to ebx and the value that was in ebx is written into eax

swap ebx, eax ; the value that was in ebx is written to eax and the value that was in eax is written into ebx

nop ; no operation - this instruction tell the processor to do nothing

Solution:

The code added has no direct change to any values so the script still functions exactly the same, it pushes a value and instantly pops it which changes nothing, it swaps values then instantly swaps them back doing nothing, then it has a nop command which literally does nothing.

This “junk” code is mainly used to pad the binary and change the code’s signature, allowing it to bypass any anti-virus scanners.

6.5 [5] Consider the following fragment:

```
legitimate code  
if data is Friday the 13th;  
    crash_computer();  
legitimate code
```

Does the presence of this logic-bomb trigger necessarily mean the fragment is a virus? Why or why not?

Solution:

No it doesn't mean the fragment is a virus.

In class we defined a virus as a piece of software that infects programs through a host, usually with a trigger of some kind. This logic bomb will not infect anything when it runs, instead it will just crash the computer. You could call it malware and it could be packaged along side a Trojan Horse or Worm, but on its own it isn't infecting any programs so it can't be categorized as a Virus.

6.6 [5] Consider the following fragment in an authentication program:

```
username = read_username();  
password = read_password();  
if username is "133t h4ck0r"  
    return ALLOW_LOGIN;  
if username and password are valid  
    return ALLOW_LOGIN  
else return DENY_LOGIN
```

What type of malicious software is this?

Solution:

This looks like a backdoor, it looks like someone on the inside that created the login system also added in some code that will allow anyone to instantly login with only the username: "133t h4ck0r".

- 6.8.** [15] Suppose you observe that your home PC is responding very slowly to information requests from the net. And then you further observe that your network gateway shows high levels of network activity even though you have closed your e-mail client, Web-browser, and other programs that access the net.

What types of malware could cause these symptoms?

Discuss how the malware might have gained access to your system.

What steps can you take to check whether this has occurred?

If you do identify malware on your PC, how can you restore it to safe operation?

Solution:

A worm or a botnet could have caused these symptoms. Worms seek to replicate themselves on other machines and a botnet is when your PC is being controlled via a remote facility to connect to other machines. Both of these would cause high network traffic like this states.

It could have gained access to your system through the following: you clicking on some phishing email or other social engineered malware trap; it being Trojan horsed into your system when you downloaded some software off the internet; or you clicked an invisible layer on a website that downloaded it (clickjacking).

To check whether this has occurred you, you can check task manager / resource monitor and see if there is any suspicious software with high network values or you can run an anti-virus check software to scan your system.

To restore your PC to safe operation your options are pretty limited, run anti-virus software and pray it gets it all, roll back to a backup of your PC if you have one, or do a full wipe on your system then reinstall your OS.

- 6.9.** [10] Suppose that while trying to access a collection of short videos on some Web site, you see a pop-up window stating that you need to install a custom codec in order to view the videos.

What threat might this pose to your computer system if you approve this installation request?

Solution:

Trojan horse threat delivered via Social Engineering.

It could be a Trojan horse by way of it being malicious software disguised as a necessary tool. It is delivered via Social Engineering by way of it appearing like its necessary, when in reality it just contains malware. Hence they got you to click and install it through deception.

The threat to your computer is that it could cause any type of bad malware to run on your PC. It could be ransomware to take hostage of your system to try and get money out of you, it could swipe all your personal information, or it could use your computer to infect others.